Environmental Product Declaration

In accordance with ISO 14025:2006 and EN 15804:2012+A2:2019/AC:2021 for:

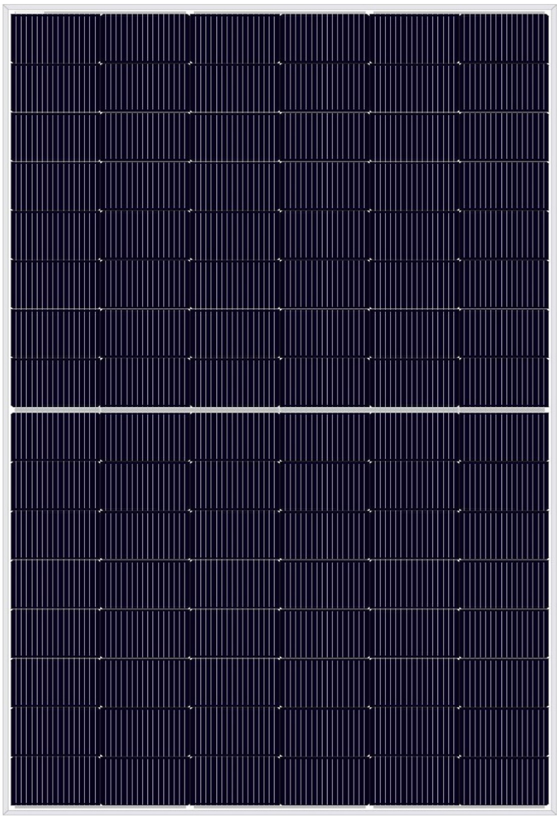
[*Photovoltaic modules*]

from

[*Tongwei Co., Ltd.*]

Programme:	The International EPD System, www.environdec.com
Programme operator:	EPD International AB
Licensee:	None
Type of EPD:	EPD of multiple products from a company
EPD registration number:	EPD-IES-0025024
Version date:	2025-08-21
Validity date:	2030-08-21 *An EPD may be updated or depublished if conditions change. To find the latest version of the EPD and to confirm its validity, see www.environdec.com*

Statement*: EPD of multiple products, based on worst case results. The list of the products is covered: TWMNH-48HExxx, TWMNH-48HWxxx and TWMNH-48HDxxx.*



GENERAL INFORMATION

Programme Information	
Programme:	The International EPD® System
Address:	EPD International AB Box 210 60 SE-100 31 Stockholm Sweden
Website:	www.environdec.com
E-mail:	support@environdec.com

Product Category Rules (PCR)
CEN standard EN 15804 serves as the Core Product Category Rules (PCR)
Product Category Rules (PCR): < *PCR 2019:14 PCR Construction products v2.0.1 issue data 2025-06-05 valid until 2030-04-07*>
PCR review was conducted by: <*Technical Committee of the International EPD® System. A full list of members available on www.environdec.com. The review panel may be contacted via info@environdec.com* *Chair of the PCR review: < Rob Rouwette >*
c-PCR, if applicable: < *c-PCR-016 Photovoltaic modules and parts thereof (adopted from EPD Norway NPCR 029 Part B Part B for photovoltaic modules used in the building and construction industry, including production of cell, wafer, ingot block, solar grade silicon, solar substrates, solar superstrates and other solar grade semiconductor materials, version 1.2, issue data 2022-03-31*>.

Third-party Verification

Independent third-party verification of the declaration and data, according to ISO 14025:2006, via:

☒ **Individual EPD verification without a pre-verified LCA/EPD tool**

Third-party verifier: <*Michael ZHU Jiang* >
Approved by: International EPD System
or
<name of certification body (including address) >
Accredited by: < *Name of accreditation body & accreditation number, where applicable*>

☐ **Individual EPD verification with a pre-verified LCA/EPD tool**
Third-party verifier: <*name, and organization of the individual verifier*>
Approved by: International EPD System
or
< name of certification body (including address) >
Accredited by: < *Name of accreditation body & accreditation number, where applicable*>

Pre-verified LCA tool or Pre-verified EPD tool: <*name and version*>
Third-party verifier, accountable for the tool verification: <*name, and organization of the individual verifier*>
Approved by: International EPD System

or
< name of certification body (including address) >
Accredited by: < *Name of accreditation body & accreditation number, where applicable*>

☐ **EPD process certification* without a pre-verified LCA/EPD tool**
Third-party verifier:
< name of certification body (including address) >
Accredited by: < *Name of accreditation body & accreditation number, where applicable*>

☐ **EPD process certification* with a pre-verified LCA/EPD tool**
Third-party verifier:
< name of certification body (including address) >
Accredited by: < *Name of accreditation body & accreditation number, where applicable*>

Pre-verified LCA tool or Pre-verified EPD tool: <*Simapro10.1* >
Third-party verifier, accountable for the tool verification:
<name, and organization of the individual verifier>
Approved by: International EPD System
or
< name of certification body (including address) >
Accredited by: < *Name of accreditation body & accreditation number, where applicable*>

☐ **Fully pre-verified EPD tool**
Fully pre-verified EPD tool: <*name and version*>
Third-party verifier, accountable for the tool and EPD verification:
<name, and organization of the individual verifier>
Approved by: International EPD System
or
< name of certification body (including address) >
Accredited by: < *Name of accreditation body & accreditation number, where applicable*>

*EPD process certification involves an accredited certification body certifying and periodically auditing the EPD process and conducting external and independent verification of EPDs that are regularly published. More information can be found in the General Programme Instructions on www.envrondec.com.

Procedure for follow-up of data during EPD validity involves third party verifier:

☐ Yes ☒ No

[Procedure for follow-up the validity of the EPD is at minimum required once a year with the aim of confirming whether the information in the EPD remains valid or if the EPD needs to be updated during its validity period. The follow-up can be organized entirely by the EPD owner or together with the original verifier via an agreement between the two parties. In both approaches, the EPD owner is responsible for the procedure being carried out. If a change that requires an update is identified, the EPD shall be re-verified by a verifier]

The EPD owner has the sole ownership, liability, and responsibility for the EPD.

EPDs within the same product category but published in different EPD programmes, may not be comparable. For two EPDs to be comparable, they shall be based on the same PCR (including the same first-digit version number) or be based on fully aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional

units); have identical scope in terms of included life-cycle stages (unless the excluded life-cycle stage is demonstrated to be insignificant); apply identical impact assessment methods (including the same version of characterisation factors); and be valid at the time of comparison.
For further information about comparability, see EN 15804 and ISO 14025.

INFORMATION ABOUT EPD OWNER

Owner of the EPD:
Tongwei Co., Ltd.

Address:
No. 588, Middle Section Tianfu Avenue, High-Tech Zone, Chengdu, China (Sichuan) Pilot Free Trade Zone

Contact:
sales@tongwei.com, +86-28-60707985

Address and contact information of the LCA practitioner commissioned by the EPD owner, if applicable: [Bella Hu; Bldg. 5, Lane 288, Kangning Road, Jing'an District, 200040, Shanghai, P.R.China]

Description of the organisation:
Tongwei Co., Ltd., controlled by Tongwei Group, is a large-scale, privately listed technology company with core businesses in green agriculture and green energy. As stated in the 2023 Annual Report, the company operates over 200 branches and subsidiaries both domestically and internationally, employing nearly 60,000 people. Tongwei boasts an annual feed production capacity exceeding 10 million tons and an annual high-purity crystalline silicon production capacity of 450,000 tons, soon to be increased to 900,000 tons with the upcoming completion of the 200,000-ton Yunnan project. Additionally, it maintains an annual solar cell production capacity of over 150GW and an annual module production capacity of 90Gw.Moreover, the company has a cumulative installed and grid-connected capacity of 4.67 GW across 56"Fishery & PV Integration" power stations. After years of rapid development, Tongwei has emerged as a leading Chinese enterprise in agricultural industrialization and a prominent producer of aquatic feed. It also ranks among the world's top producers of aquatic feed, high-purity crystalline silicon, and solar cells, while remaining a significant producer of livestock and poultry feed.

Product-related or management system-related certifications:
Product-related certifications: IEC 61215-1, IEC 61215-1-1, IEC 61215-2, IEC 61730-1, IEC 61730-2, IEC 61701, IEC 62716, IEC 60068-2-68, IEC TS 62804-1
Management system-related certifications: IS09001, IEC62941, ISO14001, IS045001

PRODUCT INFORMATION

Product name:
TWMNH-48HExxx, TWMNH-48HWxxx, TWMNH-48HDxxx

Product identification:

Serious (brand name)	Technology	Maximum output(W)	Dimensions (mm^2)	Weight (kg) excluding packaging (kg)	Weight including packaging (kg)	Module efficiency	Cell number	First year degradation	Annual average degradation
TWMNH-48HExxx	Mono-crystalline	470	1762±2*1134±2	22.4	23.539	23.5%	96	1%	0.4%

Serious (brand name)	Technology	Maximum output(W)	Dimensions (mm^2)	Weight (kg) excluding packaging (kg)	Weight including packaging (kg)	Module efficiency	Cell number	First year degradation	Annual average degradation
TWMNH-48HWxxx	Mono-crystalline	475	1762±2*1134±2	22.4	23.539	23.8%	96	1%	0.4%
TWMNH-48HDxxx	Mono-crystalline	475	1762±2*1134±2	21.9	23.039	23.8%	96	1%	0.4%

The degradation rate is based on the product specification from Tongwei Solar.

Visual representation (e.g., an image) of the product

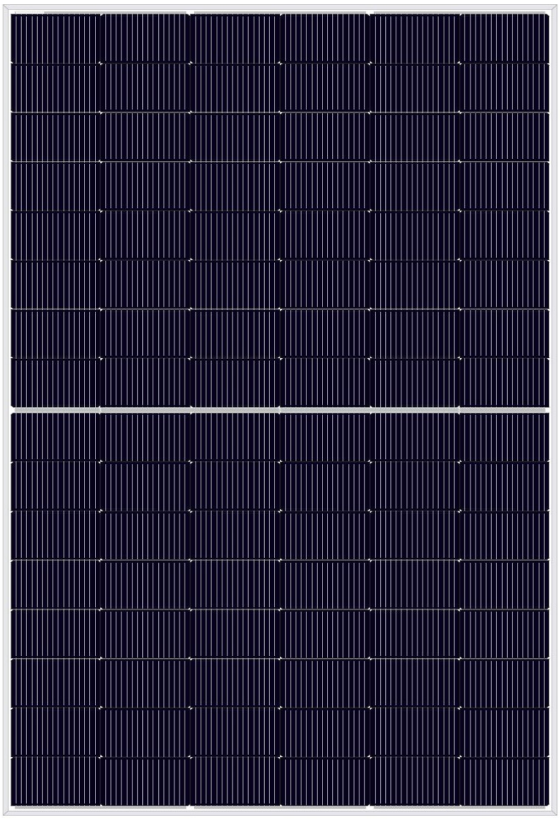


Figure 1 the product image

UN CPC code:
461 Electric motors, generators and transformers, and parts

Product description:
These PV modules are dual glass monocrystalline silicon module. The module efficiency can reach as high as 22.0-23.8% thanks to the high density interconnect technology. Modules contain multi-busbar technology for better light trapping, lower series resistance, improved current collection and enhanced reliability. The warranty period for linear power output is 30 years.

Name and location of production site(s):
No. 8, Bonded 10 Road, Economic Development Zone, Nantong City, 226017 Jiangsu Province, P.R. China

Name of manufacturer(s) (if EPD of goods) or service provider(s) (if EPD of services), if different from the EPD owner.
Tongwei Co., Ltd.

References to any relevant websites for more information or explanatory materials, if applicable.
Website: https://en.tongwei.cn/

CONTENT DECLARATION

According to the PCR, for EPD with multiple products, three options can be chosen. This EPD follows the approach for “worst case” scenario since the production volume of the three types of PV modules follows the exact same manufacturing process and supplied with a similar production volume. The TWMNH-48HExxx is used. Per the requirement of the PCR 2019:14 version 2.0.0, the content declaration is based on the weight of one unit of a product as purchased.

Product content	Mass, kg	Post-consumer recycled material, mass-% of product	Biogenic material, mass-% of product	Biogenic material, kg C/product or declared unit
Photovoltaic cell	5.45E-01	0%	0%	0
Front glass	8.83E+00	0%	0%	0
Back glass	8.91E+00	0	0	0
EVA film	1.62E+00	0	0	0
Frame	1.88E+00	0	0	0
Busbar	4.83E-02	0	0	0
Interconnection strip	1.63E-01	0	0	0
Silica gel	2.68E-01	0	0	0
Potting adhesive	2.90E-02	0	0	0
Junction box	7.90E-02	0	0	0
welding flux	1.60E-02	0	0	0
TOTAL	2.24E+01	0	0	0

Packaging materials	Mass, kg	Mass-% (versus the product)	Biogenic material, kg C/product or declared unit
wooden pallet	8.58E-01	3.83	4.05E-01
Carton	1.77E-01	0.79	7.97E-02
kraft paper	3.98E-02	0.18	1.66E-02
PET	1.27E-02	0.06	0

PE	5.20E-02	0.23	0
TOTAL	1.14E+00	5.09	5.01E-01

1 kg biogenic carbon in the product/packaging is equivalent to the uptake of 44/12 kg of CO_2.

No dangerous substances from the candidate list of SVHC for Authorisation.

LCA INFORMATION

Functional unit:

1 Wp of manufactured photovoltaic module, from cradle-to-grave, with activities needed for a study period for a defined reference service life (⩾80% of the labelled power output). Since no third-party report is available, a standard reference service life of 25 years for ⩾80% of the labelled power output is applied.

Table 1 The conversion factor for functional unit to the declared unit

PV modules	Cell dimension (m2)	Value(W/m2)
TWMNH-48HExxx	1.998	235
TWMNH-48HWxxx	1.998	238
TWMNH-48HDxxx	1.998	238

Reference service life:

25 years

Time representativeness:

Data collection period: 2024-05-01 to 2025-04-30

Steps were taken to ensure that the LCI data were reliable and representative. The data type used is clearly stated in the Inventory analysis, measured or calculated from primary sources or whether data are from the LCI databases. In this study, the data quality requirements were as follows:

Specific data of the considered system (such as material or energy flows that enter the production system). These data were calculated and submitted by Tongwei.

Generic data related to the life cycle impacts the material or energy flows that enter the production system. These data were sourced from the databases in SimaPro 10.1.

Geographical scope:

China for A1-A3, and Europe for A4, A5, B2, and C1-C4.

Database(s) and LCA software used:

Database: Ecoinvent 3.10.1, Ecoinvent 3 – allocation, cut-off by classification – unit

LCA Software: Simapro 10.1

Description of system boundaries:

The system boundary considered in this LCA study is “cradle to gate with options, modules C1-C4, module D with optional modules (A1-A3 + A4 +A5+ B2 + C + D)”.

A1-A3: Product stage (raw material acquisition, transport to manufacturing site and manufacturing)

A4: Transport to user site

A5: Installation

B2: Maintenance of product

C1-C4: End-of-life stage (deconstruction, transport, waste processing and disposal)

D: Reuse, recovery and/or recycling potentials

A1 Raw materials extraction

Raw materials extraction includes materials needed to produce ingot, wafer, cell and PV module. Ingot, wafer and cell can be regarded as the intermediate products along the PV module production line. The raw materials extraction for the three types Tongwei PV modules are similar. The PV cells are manufactured by TW Solar Technology Co., Ltd. The wafer is sourced from the Ecoinvent datasets “silicon production, single crystal, Czochralski process, photovoltaics RoW”.

A2 Raw materials transport

Concerning the raw material transportation, all the raw materials are sourced from domestic suppliers and are transported by truck, EURO5 is used for modelling in this study. The 16-32t transportation type scenario is assumed. The study applies an aggregated approach for the raw materials transportation summarizing all the transport data through multiplying the weight and the transportation distance.

A3 Module Assembly

For A3 stage, the electricity consumption mix is based on the electricity generation profile for the Jiangsu Province, which is the dedicated location of the manufacturing site of the module assembly. The details of the electricity mix are shown in Figure 1 below. Per the PCR, a grid loss of 3.07% in Jiangsu province is considered to convert the production mix to the consumption. The electricity emission factor of GWP-GHG is 0.909kgCO_{2e}/kWh.

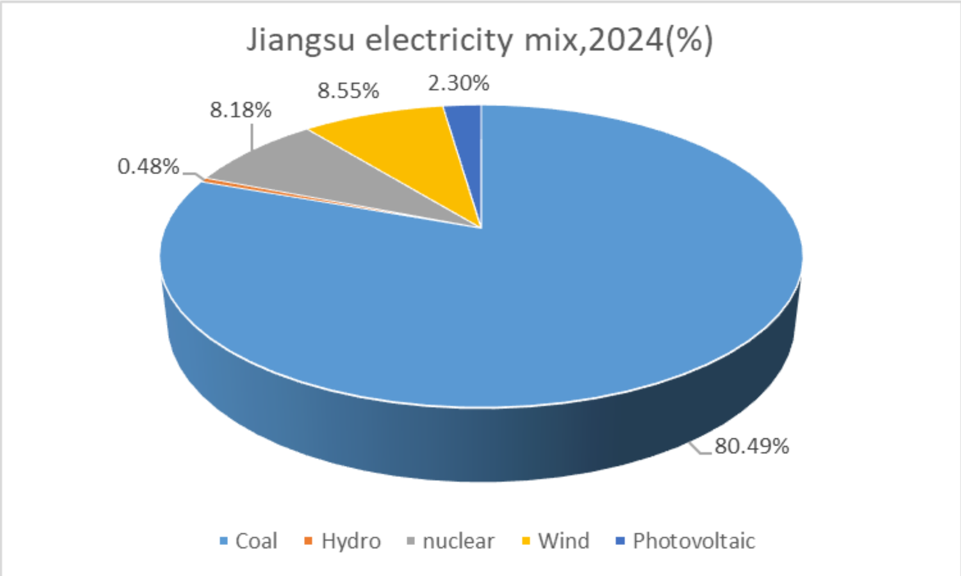


Figure 2 the electricity production mix for Jiangsu province at 2024

Table 2 Energy sources for the product manufacturing processes

	TWMNH-48HExxx	TWMNH-48HWxxx	TWMNH-48HDxxx
Manufacturing site location	Nantong, Jiangsu		
Underlying ecoinvent dataset	Electricity production in Jiangsu Province		
GWP-GHG (kg CO_2 eq. / kWh)	0.909		

A4 Product distribution

The distribution scenario(A4) is based on the information provided by the suppliers. The product is firstly transported to the Shanghai Port at the distance of 91km by lorry. Then, the product is transported from Shanghai Port to Civitavecchia Port by 15712km through the container ship. After then, the product is unloaded to the storage site by the distance of 300km. Finally, the distance between the storage site to the final consumers is assumed to be 500km for European domestic construction site according to the NPCR 029 Part B version 1.2.

A5 Installation

The packaging materials of the PV modules are mostly wooden pallet and paper, and are assumed to be recycled. The transport distance for the packaging materials to the recycling site is assumed to be 50km. The scaling method is clearly listed in Table 3. Other materials including the mounting system, cables, inverts are not considered based on the requirements listed in the NPCR 029 Part B version 1.2.

Table 3 Energy inputs for PV module installation

Construction consumption process (per kWp capacity)		
Electricity	36.033 kWh electricity for 570kWp as in the Ecoinvent dataset (“Photovoltaic plant, 570kWp, multi-Si, on open ground {GLO}| photovoltaic plant construction, 570kWp, multi-Si, on open ground | Cut-off, U”)	0.06 kWh/kWp is applied
Diesel	7673MJ diesel for 570kWp as in the Ecoinvent dataset(“Photovoltaic plant, 570kWp, multi-Si, on open ground {GLO}| photovoltaic plant construction, 570kWp, multi-Si, on open ground | Cut-off, U”)	13.4MJ/kWp is applied

B1-B7 modules

B1 -use of the installed product, service, or appliance There isn’t any energy and material consumption in this stage in the site.
B2-maintenance of product The only maintenance for PV panels is cleaning. It is assumed to be cleaned once per month with an application rate of 0.76L water per m2 PV panel.
No require (B3), replacement (B4) or refurbishment (B5) are needed for the PV panels.
B6 – operational energy. The product doesn’t consume energy during the whole service life. It produces the energy.
B7 – operational water use This operational stage, there isn’t any water consumption.
Electricity generation can be calculated according to the following mechanism. The site information for the simulation has the following characteristics.

Table 4 Power station information for simulation (Albacete is selected as the reference point)

Item	Value
Location	Rome, Italy
Peak power of the plant	1MW
Latitude	41.897°N
Longitude:	013.482°W
Altitude	37m
Nominal solar irradiance	1523 kWh/m2/year

Energy production in the first year of operation:

$$E1 = S_{rad} * A * PR * y * (1- deg_1) \text{ ----------------------------------- } (1)$$

E_1--- Energy produced in the first year of operation, kWh/year
S_{rad}---Site specific annual average solar radiation on module (shadings not included), kWh/kWp/year. The annual radiation must take into consideration the specific inclination (slope, tilt) and orientation.
A --- Area of module, from functional unit (FU), m2 (stated in the EPD).
Y --- Module yield: electrical power, kWp for standard test conditions (STC) of the module divided by the area of the module (stated in the EPD).
PR--- Performance ratio, coefficient for losses. 86% in our case considering the default settings in the PVGIS software.
deg_1--- first year degradation rate, in our case 1% according to the product specification

Energy production n year of operation:

$$E_n = E_1 * (1-deg)^{n-1} \quad (2)$$

Energy production over reference service life of module, assuming linear annual degradation:

$$E_{RSL} = E_1 * \left(1 + \sum_{n=1}^{RSL-1}(1-deg)^n\right) \quad (3)$$

Table 5 Total electricity generation over RSL

Serious (brand name)	Maximum power output range(W)	PR ratio	deg-first year	deg-after first year	E_1/kWh	E_{RSL}/kwh
TWMNH-48HExxx	470	86%	1%	0.4%	609.44	14526.63
TWMNH-48HWxxx	475	86%	1%	0.4%	615.92	14681.17
TWMNH-48HDxxx	475	86%	1%	0.4%	615.92	14681.17

C1-C4 modules

De-construction (C1) of the PV plant during the disposal stage is assumed mainly consuming electricity, and the electricity consumption is assumed the same as the construction stage (A5), 50km transportation distance from plant site to waste treatment site (C2) is assumed to be 50km according to the NPCR 029 Part B version 1.2. For the C3 phase, since there is lack of existing data of recycling rate for PV module, this study refers to legal requirements issued by Waste Electrical and Electronic Equipment (WEEE). In 2012/19/EU-Article 11 & ANNEX V, the required collection rate for waste PV module is 85%. Therefore, 15% of waste PV module end up with waste disposal through landfill. A specific electricity 0.0556kWh/kg and 0.0324MJ/kg diesel consumption is referenced to dissemble and sort the collected PV modules. The final disposal scenario for C4 is based on the following Table 6.

Table 6 Waste treatment scenario for PV modules

PV components	Materials	Recycling	Landfill	Incineration
PV cells	Silicon	80%	20%	0%
	Silver bar line	90%	10%	0%
Solar glass	Glass	85%	15%	0%
PET	PET	0%	0%	100%
Aluminium Frame	Aluminium alloy	94%	6%	0%
Interconnection and busbar string	Copper	63%	37%	0%
	Pb	93%	7%	0%
	Tin	90%	10%	0%
Junction box	Bronze	63%	37%	0%
	Plastics	0%	0%	100%
Chemicals	Adhesive	0%	0%	100%
EVA	EVA	0%	0%	100%

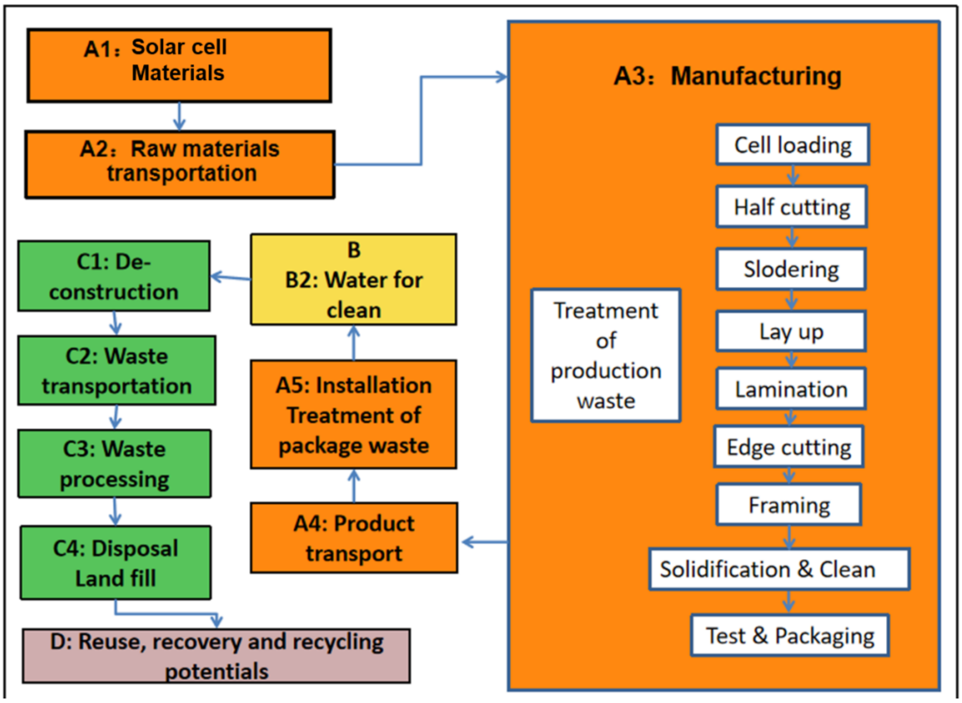
Module D

According to the PCR, Module D assesses the impact of the net flows of recovered materials (recycled or reused) from the life cycle stages A to C, the net flow can be described by the difference between $M_{MR\ in}$ and $M_{MR\ out}$, taking the material yield (here designated with Y) into account.

$$Netflow = \Sigma\ (M_{MR\ out} - Y \cdot M_{MR\ in})$$

In this LCA study, no secondary material was used in the production stage, so the $M_{MR\ in}$ is zero. The *Netflow* is Aluminium, copper, silver scraps and metallurgical silicon and others recovered from the PV modules. Meanwhile, the incineration considers electricity and heat exports.

Process flow diagram:



More information:

Cut-off criteria

No specific materials have been cut-off in this specific LCA. All materials provided by the manufacturer are properly modelled. However, the following steps/stages are not included in the system boundary for the reason that the elements below are considered irrelevant or can be omitted according to the PCR

- Production and disposal of the infrastructure and capital equipment (buildings, machines, transport media, roads, etc.) during products manufacturing, installation, and maintenance;
- The load and benefit of recycling waste solar module as well as waste equipment from solar plant are excluded from the analysis;
- The packaging for ingot, wafer and solar cell is reused internally and its impact was excluded from the system;
- Storage phases and sales of PV modules;
- Product losses due to abnormal damage such as natural disasters or fire accidents. These losses would mostly be accidental;
- The recycling process of defective products as it is reused internally for the manufacturing process;
- Handling operations at the distribution center and retail outlet due to small contribution and negligible impact.

Data quality assessment

The overall data quality is summerized according to EN15941 Annex C, The EPD is based on data collected by Enterprises-Tongwei Co. Ltd. from its Nantong manufacturing plant over one year from May 2024. The EPD is representative of the production of TWMNH-48HExxx, TWMNH-48HWxxx and

TWMNH-48HDxxx at Tongwei Enterprise's Nantong site and for all the specific PV modules produced Tongwei Enterprise. The use and end-of-life stage of the EPD covers the situation in the Europe. Enterprises-Tongwei uses technology TNC (Tongwei N-passivated contact cell) using primary material at Nantong manufacturing plant site. The Nantong manufacturing plant site is selected since the three type PV modules are exclusively produced from this area. the The EPD uses background data from the database Ecoinvent 3.10.1. The quality of the generic and producer-specific data used for the EPD in terms of its time, geography and technology representativeness are evaluated using EN 15804:2012+A2:2019, Annex E, E1. The primary data are found to be range from good to very good for its quality while the data quality of generic data can also be regarded as fair to good. The EPD uses local electricity mix to improve its geographical representativeness though the manufacturing electricity consumption only contributes to a small fraction of 2.1% in climate change-total category. The relevant data assessed included no other poor or very poor data.

The processes contributing more than 10% of the total GWP-GHG for A1-A3 is declared in the following:

Process	Source type	Source	Reference year	Data category	Share of processes relative to A1-A3 in GWP-GHG
Production of Photovoltaic cells	Database	Ecoinvent v.3.10	2024	Secondary data	78%

Production of the photovoltaic cells is the only process contributing to more than 10% of total GWP-GHG in A1-A3. Next to the processes of the photovoltaic production is the solar glass production and aluminium frame production where 8.04% and 5.61% shares are recorded.

Allocation

Since the three modules are produced from the same production line. Therefore, a multi-output allocation strategy is applied for the A3 phase to the specific PV modules. In this study, the parameter of “nominal power output” is referred as the allocation reference to partition the environmental impact for the PV modules.

The allocation strategy for the EoL process per PCR follows the same strategy listed in the EN15804. Thus, the “cut-off” strategy is applied. This scenario allocates the entire environmental impacts of waste treatment procedures (from deconstruction to the waste processing) to the producer. The recycled materials, on the other hand, are burden-free. An important note is that when materials have reached a so-called “end-of-waste” state, the coverage of the waste processing is thus terminated. Any inputs/flows related to refine gross recycled materials for actual applications are beyond the product system boundary and is accounted in Module D.

Key assumptions

Categories	Items	Type	Assumptions
Transportation stage (A2, and A4)	Transportation vehicle type	Assumption	For transport without detailed information, EURO 5 type vehicle with 16-32 ton capacity is used
Installation stage (A5)	PV module and infrastructures	Assumption	No construction waste is considered Packaging materials for PV modules are assumed be transported at a distance 100km Energy consumption for the construction process is sourced from the Ecoinvent dataset “electric installation for 570kWp module, open

			ground{GLO}|market for photovoltaics, electric installation for 570kWp module, open ground”
Use & Maintenance(B)	Use	Scenario according to NPCR 029 Part B version 1.2	The use stage requires no energy and materials inputs, and has no emissions.
	Cleaning maintenance	Scenario according to NPCR 029 Part B version 1.2	Water used for cleaning the PV panels is assumed 0.76L/m2 for 12 times per year
	Repair Replacement	Scenario according to Peer reviewed literature	No replacement for the module as the module has RSL>25 years. No operational water and energy are needed for PV module
End-of-life (C1-C4)	Inputs for de-installation	Assumption	The de-construction of PV plant is assumed to be consuming the same energy as the installation stage
	Waste transport distance	Scenario according to the NPCR 029 Part B version 1.2	Waste transportation distance from the de-installation plant to the waste treatment facilities is assumed to be 50 km according to the NPCR 029 Part B version 1.2
	Inputs for disassembly the PV modules	Scenario according to the Peer reviewed Literature	Report IEA-PVPS T12-19:2020
	Waste treatment scenarios for dissembled PV modules	Scenario according to the Peer reviewed Literature	December 2020 is referenced for the energy and diesel consumption to treat and dissemble the PV modules.
Module D	Exported energy	Assumption	For incineration with the energy recovery, net energy production: 3.93MJ/kg electric energy and 7.66MJ/kg thermal energy.

Modules declared, geographical scope, share of specific data (in GWP-GHG results) and data variation (in GWP-GHG results):

	Product stage			Construction process stage		Use stage							End of life stage				Resource recovery stage
	Raw material supply	Transport	Manufacturing	Transport	Construction installation	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
Module	**A1**	**A2**	**A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
Modules declared	x	x	x	x	x	ND	x	ND	ND	ND	ND	ND	x	x	x	x	x
Geography	CN	CN	CN	EUR	EUR		EUR						EUR	EUR	EUR	EUR	EUR
Specific data used	3.19%					-	-	-	-	-	-	-	-	-	-	-	-
Variation – products	<10%					-	-	-	-	-	-	-	-	-	-	-	-
Variation – sites	0%					-	-	-	-	-	-	-	-	-	-	-	-

ENVIRONMENTAL PERFORMANCE

LCA results of the product(s) - main environmental performance results based on the worst-case scenario.

The Life Cycle Assessment study has been performed in accordance with the requirements of EN15804+A2:2019 (version 1.02), PCR 2019:14 (v2.0.1 issue data 2025-06-05 valid until 2030-04-07) and JRC characterization factors EF 3.1.

Mandatory impact category indicators according to EN 15804

Results per functional unit										
Indicator	**Unit**	**A1-A3**	**A4**	**A5**	**B2**	**C1**	**C2**	**C3**	**C4**	**D**
GWP-total	kg CO_2 eq.	4.93E-01	1.79E-02	5.59E-03	1.24E-03	1.39E-03	4.62E-04	1.06E-03	1.08E-02	-9.50E-02
GWP-fossil	kg CO_2 eq.	4.95E-01	1.79E-02	1.62E-03	1.24E-03	1.39E-03	4.61E-04	1.05E-03	1.08E-02	-9.47E-02
GWP-biogenic	kg CO_2 eq.	-2.57E-03	2.86E-06	3.97E-03	1.12E-06	1.24E-07	8.26E-08	4.37E-06	4.76E-07	-1.88E-04
GWP-luluc	kg CO_2 eq.	5.37E-04	8.12E-06	1.29E-07	1.02E-06	1.20E-07	1.51E-07	1.64E-07	5.88E-08	-2.34E-05
ODP	kg CFC 11 eq.	2.82E-03	2.70E-04	1.24E-05	6.41E-06	1.22E-05	1.44E-06	4.66E-06	3.08E-06	-8.42E-04
AP	mol H^+ eq.	2.19E-04	1.02E-06	4.25E-08	4.18E-07	4.02E-08	3.08E-08	1.47E-07	6.72E-08	-4.33E-05
EP-freshwater	kg P eq.	5.26E-04	6.95E-05	5.77E-06	1.30E-06	5.66E-06	4.86E-07	1.25E-06	1.98E-06	-1.16E-04
EP-marine	kg N eq.	5.48E-03	7.70E-04	6.25E-05	1.34E-05	6.20E-05	5.29E-06	1.36E-05	1.59E-05	-1.29E-03
EP-terrestrial	mol N eq.	1.99E-03	2.23E-04	1.87E-05	4.09E-06	1.85E-05	2.26E-06	4.93E-06	4.08E-06	-3.87E-04
POCP	kg NMVOC eq.	2.41E-08	2.53E-10	2.22E-11	2.24E-10	2.18E-11	9.17E-12	2.24E-11	5.35E-12	-4.47E-10
ADP-minerals& metals	kg Sb eq.	1.81E-05	3.92E-08	6.16E-10	5.51E-09	5.34E-10	1.47E-09	1.78E-09	4.48E-10	-1.05E-05
ADP-fossil	MJ	6.29E+00	2.36E-01	1.87E-02	1.48E-02	1.84E-02	6.48E-03	1.71E-02	2.92E-03	-8.84E-01
WDP	m^3	3.95E-01	8.48E-04	4.16E-05	4.14E-02	4.07E-05	2.65E-05	-5.11E-04	-7.30E-04	-1.25E-02
Acronyms	GWP-fossil = Global Warming Potential fossil fuels; GWP-biogenic = Global Warming Potential biogenic; GWP-luluc = Global Warming Potential land use and land use change; ODP = Depletion potential of the stratospheric ozone layer; AP = Acidification potential, Accumulated Exceedance; EP-freshwater = Eutrophication potential, fraction of nutrients reaching freshwater end compartment; EP-marine = Eutrophication potential, fraction of nutrients reaching marine end compartment; EP-terrestrial = Eutrophication potential, Accumulated Exceedance; POCP = Formation potential of tropospheric ozone; ADP-minerals&metals = Abiotic depletion potential for non-fossil resources; ADP-fossil = Abiotic depletion for fossil resources potential; WDP = Water (user) deprivation potential, deprivation-weighted water consumption									

** Disclaimer:*
The estimated impact results are only relative statements, which do not indicate the endpoints of the impact categories, exceeding threshold values, safety margins and/or risks.
If the EPD covers the end-of-life stage: "The results of the end-of-life stage (modules C1-C4) should be considered when using the results of the product stage (modules A1-A3)" For services, "A1-A3" shall be replaced by "A1-A5

If results based on an old EF version is used to develop an EPD, the EPD shall include a statement that clarifies that an EPD based on an old EF version has been used as a data source, and that this was assessed to yield identical or conservative results compared to fully using the current EF version.

Additional mandatory and voluntary impact category indicators

Results per functional unit										
Indicator	**Unit**	**A1-A3**	**A4**	**A5**	**B2**	**C1**	**C2**	**C3**	**C4**	**D**
GWP-GHG[1]	kg CO_2 eq.	4.97E-01	1.79E-02	1.69E-03	1.24E-03	1.39E-03	4.62E-04	1.06E-03	1.08E-02	-9.50E-02
Additional voluntary indicators e.g. the voluntary indicators from EN 15804 or the global indicators according to ISO 21930:2017										

**The biogenic carbon leaving the product system in module A5 have been balanced out already in modules A1-A3.*

Resource use indicators

Results per functional unit										
Indicator	**Unit**	**A1-A3**	**A4**	**A5**	**B2**	**C1**	**C2**	**C3**	**C4**	**D**
PERE	MJ	1.06E+00	2.44E-03	1.14E-04	1.37E-03	1.07E-04	1.06E-04	4.67E-03	8.03E-05	-6.54E-02
PERM	MJ	3.06E-02	0.00E+00	-3.03E-02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ	1.09E+00	2.44E-03	-3.02E-02	1.37E-03	1.07E-04	1.06E-04	4.67E-03	8.03E-05	-6.54E-02
PENRE	MJ	6.15E+00	2.36E-01	1.87E-02	1.48E-02	1.84E-02	6.48E-03	1.71E-02	2.92E-03	-8.84E-01
PENRM	MJ	1.40E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	-1.39E-01	0.00E+00
PENRT	MJ	6.29E+00	2.36E-01	1.87E-02	1.48E-02	1.84E-02	6.48E-03	1.71E-02	-1.36E-01	-8.84E-01
SM	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	m^3	1.46E-02	2.56E-05	1.36E-06	9.65E-04	1.34E-06	8.87E-07	-1.00E-05	-1.38E-05	-3.26E-04
Acronyms	PERE = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; PERM = Use of renewable primary energy resources used as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy resources used as raw materials; PENRT = Total use of non-renewable primary energy re-sources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water									

Waste indicators

[1] This indicator accounts for all greenhouse gases except biogenic carbon dioxide uptake and emissions and biogenic carbon stored in the product. As such, the indicator is identical to GWP-total except that the CF for biogenic CO_2 is set to zero.

Results per functional unit										
Indicator	**Unit**	**A1-A3**	**A4**	**A5**	**B2**	**C1**	**C2**	**C3**	**C4**	**D**
Hazardous waste disposed	kg	5.49E-04	4.81E-06	2.69E-06	8.19E-07	1.66E-07	1.60E-07	7.61E-07	2.07E-04	-2.85E-05
Non-hazardous waste disposed	kg	2.06E-02	6.61E-03	2.61E-04	8.99E-05	1.15E-05	3.07E-04	1.43E-02	1.08E-02	-1.92E-03
Radioactive waste disposed	kg	1.06E-05	3.86E-08	2.07E-09	2.52E-08	1.96E-09	2.06E-09	2.76E-08	1.22E-09	-4.47E-07

Output flow indicators

Results per functional unit										
Indicator	**Unit**	**A1-A3**	**A4**	**A5**	**B2**	**C1**	**C2**	**C3**	**C4**	**D**
Components for re-use	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Material for recycling	kg	0.00E+00	0.00E+00	4.22E-04	0.00E+00	0.00E+00	0.00E+00	3.16E-02	0.00E+00	0.00E+00
Materials for energy recovery	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Exported energy, electricity	MJ	1.77E-05	0.00E+00	3.06E-04	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.60E-02	0.00E+00
Exported energy, thermal	MJ	2.39E-04	0.00E+00	6.01E-04	0.00E+00	0.00E+00	0.00E+00	0.00E+00	3.08E-02	0.00E+00

ADDITIONAL ENVIRONMENTAL INFORMATION

Deviations for max-min results among the three modules

Impact category	Unit	Max(A-C)	Min(A-C)	Deviation (Max-Min)/Max
Climate change	kg CO2 eq	5.31E-01	5.14E-01	3%
Climate change - Fossil	kg CO2 eq	5.29E-01	5.12E-01	3%
Climate change - Biogenic	kg CO2 eq	1.42E-03	1.35E-03	5%
Climate change - Land use and LU change	kg CO2 eq	5.47E-04	5.28E-04	3%
Acidification	mol H+ eq	3.13E-03	3.03E-03	3%
Eutrophication, freshwater	kg P eq	2.21E-04	2.14E-04	3%
Eutrophication, marine	kg N eq	6.12E-04	5.93E-04	3%

Eutrophication, terrestrial	mol N eq	6.43E-03	6.22E-03	3%
Photochemical ozone formation	kg NMVOC eq	2.26E-03	2.19E-03	3%
Ozone depletion	kg CFC11 eq	2.47E-08	2.39E-08	3%
Resource use, minerals and metals	kg Sb eq	1.81E-05	1.76E-05	3%
Resource use, fossils	MJ	6.61E+00	6.40E+00	3%
Water use	m3 depriv.	4.36E-01	4.22E-01	3%
GWP-GHG	kg CO2 eq	5.31E-01	5.14E-01	3%

The three scenarios of the End of life (100% landfill, 100% incineration, 100% recycling) have been declared in the EPD.

The 100% landfill scenario

Results per functional unit						
Indicator	Unit	C1	C2	C3	C4	D
Climate change	kg CO2 eq	1.39E-03	4.62E-04	1.15E-03	1.33E-02	0.00E+00
Climate change - Fossil	kg CO2 eq	1.39E-03	4.61E-04	1.14E-03	1.33E-02	0.00E+00
Climate change - Biogenic	kg CO2 eq	1.24E-07	8.26E-08	4.56E-06	2.88E-06	0.00E+00
Climate change - Land use and LU change	kg CO2 eq	1.20E-07	1.51E-07	1.42E-07	2.87E-07	0.00E+00
Acidification	mol H+ eq	1.22E-05	1.44E-06	4.83E-06	6.58E-06	0.00E+00
Eutrophication, freshwater	kg P eq	4.02E-08	3.08E-08	1.64E-07	1.24E-07	0.00E+00
Eutrophication, marine	kg N eq	5.66E-06	4.86E-07	1.22E-06	3.49E-06	0.00E+00
Eutrophication, terrestrial	mol N eq	6.20E-05	5.29E-06	1.33E-05	3.11E-05	0.00E+00
Photochemical ozone formation	kg NMVOC eq	1.85E-05	2.26E-06	4.87E-06	9.08E-06	0.00E+00
Ozone depletion	kg CFC11 eq	2.18E-11	9.17E-12	2.39E-11	1.78E-11	0.00E+00
Resource use, minerals and metals	kg Sb eq	5.34E-10	1.47E-09	1.89E-09	1.55E-09	0.00E+00
Resource use, fossils	MJ	1.84E-02	6.48E-03	1.81E-02	1.30E-02	0.00E+00
Water use	m3 depriv.	4.07E-05	2.65E-05	7.16E-04	-6.16E-03	0.00E+00
GWP-GHG	kg CO2 eq	1.39E-03	4.62E-04	1.15E-03	1.33E-02	0.00E+00
PERE	MJ	1.07E-04	1.06E-04	5.47E-03	3.50E-04	0.00E+00
PERM	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ	1.07E-04	1.06E-04	5.47E-03	3.50E-04	0.00E+00
PENRE	MJ	1.84E-02	6.48E-03	1.81E-02	1.30E-02	0.00E+00
PENRM	MJ	0.00E+00	0.00E+00	0.00E+00	-1.40E-01	0.00E+00
PENRT	MJ	1.84E-02	6.48E-03	1.81E-02	-1.27E-01	0.00E+00
SM	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	m3	1.34E-06	8.87E-07	1.89E-05	-1.57E-04	0.00E+00
HWD	kg	1.66E-07	1.60E-07	3.55E-07	3.34E-04	0.00E+00
NHWD	kg	1.15E-05	3.07E-04	3.90E-05	9.13E-02	0.00E+00
RWD	kg	1.96E-09	2.06E-09	3.20E-08	5.09E-09	0.00E+00

CRU	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
MFR	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
MER	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
EEE	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
EET	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

The 100% incineration scenario

Results per functional unit						
Indicator	Unit	C1	C2	C3	C4	D
Climate change	kg CO2 eq	1.39E-03	4.62E-04	1.15E-03	1.39E-02	-7.64E-03
Climate change - Fossil	kg CO2 eq	1.39E-03	4.61E-04	1.14E-03	1.39E-02	-7.64E-03
Climate change - Biogenic	kg CO2 eq	1.24E-07	8.26E-08	4.56E-06	1.09E-06	-2.80E-06
Climate change - Land use and LU change	kg CO2 eq	1.20E-07	1.51E-07	1.42E-07	2.29E-06	-5.54E-07
Acidification	mol H+ eq	1.22E-05	1.44E-06	4.83E-06	9.37E-06	-2.65E-05
Eutrophication, freshwater	kg P eq	4.02E-08	3.08E-08	1.64E-07	2.19E-07	-1.47E-06
Eutrophication, marine	kg N eq	5.66E-06	4.86E-07	1.22E-06	4.45E-06	-3.02E-06
Eutrophication, terrestrial	mol N eq	6.20E-05	5.29E-06	1.33E-05	4.16E-05	-3.05E-05
Photochemical ozone formation	kg NMVOC eq	1.85E-05	2.26E-06	4.87E-06	1.29E-05	-2.93E-05
Ozone depletion	kg CFC11 eq	2.18E-11	9.17E-12	2.39E-11	2.68E-11	-9.27E-11
Resource use, minerals and metals	kg Sb eq	5.34E-10	1.47E-09	1.89E-09	3.72E-09	-4.88E-09
Resource use, fossils	MJ	1.84E-02	6.48E-03	1.81E-02	2.16E-02	-8.91E-02
Water use	m3 depriv.	4.07E-05	2.65E-05	7.16E-04	5.94E-04	-3.00E-04
GWP-GHG	kg CO2 eq	3.46E-10	3.62E-11	5.69E-11	2.01E-10	-3.72E-10
PERE	MJ	7.98E-06	8.31E-06	1.21E-04	2.40E-05	-1.96E-05
PERM	MJ	2.56E-03	1.74E-03	2.24E-03	1.11E-01	-1.68E-02
PERT	MJ	5.34E-12	3.22E-12	2.74E-12	1.70E-11	-1.76E-11
PENRE	MJ	2.27E-12	4.04E-12	4.45E-12	1.79E-10	-3.50E-11
PENRM	MJ	1.26E-03	3.85E-03	2.98E-03	2.49E-02	-1.64E-02
PENRT	MJ	1.39E-03	4.62E-04	1.15E-03	1.39E-02	-7.64E-03
SM	kg	1.07E-04	1.06E-04	5.47E-03	4.04E-04	-1.90E-03
RSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF	MJ	1.07E-04	1.06E-04	5.47E-03	4.04E-04	-1.90E-03
FW	m3	1.84E-02	6.48E-03	1.81E-02	2.16E-02	-8.91E-02
HWD	kg	0.00E+00	0.00E+00	0.00E+00	-1.40E-01	0.00E+00
NHWD	kg	1.84E-02	6.48E-03	1.81E-02	-1.18E-01	-8.91E-02
RWD	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
CRU	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
MFR	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
MER	kg	1.34E-06	8.87E-07	1.89E-05	2.01E-05	-9.15E-06
EEE	MJ	1.66E-07	1.60E-07	3.55E-07	4.92E-02	-1.12E-06
EET	MJ	1.15E-05	3.07E-04	3.90E-05	9.49E-04	-8.17E-05

The 100% recycling scenario

Results per functional unit						
Indicator	Unit	C1	C2	C3	C4	D
Climate change	kg CO2 eq	1.39E-03	4.62E-04	1.15E-03	0.00E+00	-1.33E-01

Climate change - Fossil	kg CO2 eq	1.39E-03	4.61E-04	1.14E-03	0.00E+00	-1.32E-01
Climate change - Biogenic	kg CO2 eq	1.24E-07	8.26E-08	4.56E-06	0.00E+00	-2.91E-04
Climate change - Land use and LU change	kg CO2 eq	1.20E-07	1.51E-07	1.42E-07	0.00E+00	-4.04E-05
Acidification	mol H+ eq	1.22E-05	1.44E-06	4.83E-06	0.00E+00	-1.13E-03
Eutrophication, freshwater	kg P eq	4.02E-08	3.08E-08	1.64E-07	0.00E+00	-6.05E-05
Eutrophication, marine	kg N eq	5.66E-06	4.86E-07	1.22E-06	0.00E+00	-1.62E-04
Eutrophication, terrestrial	mol N eq	6.20E-05	5.29E-06	1.33E-05	0.00E+00	-1.80E-03
Photochemical ozone formation	kg NMVOC eq	1.85E-05	2.26E-06	4.87E-06	0.00E+00	-5.48E-04
Ozone depletion	kg CFC11 eq	2.18E-11	9.17E-12	2.39E-11	0.00E+00	-8.64E-10
Resource use, minerals and metals	kg Sb eq	5.34E-10	1.47E-09	1.89E-09	0.00E+00	-1.51E-05
Resource use, fossils	MJ	1.84E-02	6.48E-03	1.81E-02	0.00E+00	-1.44E+00
Water use	m3 depriv.	4.07E-05	2.65E-05	7.16E-04	0.00E+00	-2.06E-02
GWP-GHG	kg CO2 eq	1.39E-03	4.62E-04	1.15E-03	0.00E+00	-1.33E-01
PERE	MJ	1.07E-04	1.06E-04	5.47E-03	0.00E+00	-9.87E-02
PERM	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ	1.07E-04	1.06E-04	5.47E-03	0.00E+00	-9.87E-02
PENRE	MJ	1.84E-02	6.48E-03	1.81E-02	0.00E+00	-1.44E+00
PENRM	MJ	0.00E+00	0.00E+00	0.00E+00	-1.40E-01	0.00E+00
PENRT	MJ	1.84E-02	6.48E-03	1.81E-02	-1.40E-01	-1.44E+00
SM	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	m3	1.34E-06	8.87E-07	1.89E-05	0.00E+00	-5.44E-04
HWD	kg	1.66E-07	1.60E-07	3.55E-07	0.00E+00	-4.69E-05
NHWD	kg	1.15E-05	3.07E-04	3.90E-05	0.00E+00	-3.11E-03
RWD	kg	1.96E-09	2.06E-09	3.20E-08	0.00E+00	-7.94E-07
CRU	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
MFR	kg	0.00E+00	0.00E+00	4.77E-02	4.77E-02	0.00E+00
MER	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
EEE	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
EET	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

Tongwei Co. Ltd. and its PV module products obtain the following comprehensive products and organization certification.

For products:

IEC 61215-1, IEC 61215-1-1, IEC 61215-2, IEC 61730-1, IEC 61730-2: Terrestrial photovoltaic (PV) modules – Design qualification and type approval

IEC 61730-1, IEC 61730-2: Photovoltaic (PV) module safety qualification

IEC 62716: Photovoltaic (PV) modules - Ammonia corrosion testing

IEC 61701: Photovoltaic (PV) modules - Salt mist corrosion testing

IEC 60068-2-68: environmental testing - part 2: Tests - Test L: Dust and Sand

IEC TS 62804-1: Photovoltaic (PV) modules – Test methods for the detection of potential-induced degradation - Part 1: Crystalline silicon

For the organization management:

ISO 9001: Quality Management System
IEC 62941: Terrestrial photovoltaic (PV) modules - Quality system for PV module manufacturing

For EHS and energy management:
ISO14001: Environmental Management System
ISO45001: Occupational Health and Safety Management System
ISO50001: Energy Management
IEC TS 62994:2019: Photovoltaic (PV) modules through the life cycle – Environmental health and safety (EH&S) risk assessment-General principles and nomenclature

ADDITIONAL SOCIAL AND ECONOMIC INFORMATION

None

INFORMATION RELATED TO SECTOR EPD

None

ABBREVIATIONS

Abbreviation	Definition
General Abbreviations	
EN	European Norm (Standard)
EPD	Environmental Product Declaration
EF	Environmental Footprint
GPI	General Programme Instructions
ISO	International Organization for Standardization
LCA	Life Cycle Assessment
PCR	Product Category Rules
c-PCR	Complementary Product Category Rules
CEN	European Committee for Standardization
CLC	Co-location centre
CPC	Central product classification
GHS	Globally harmonized system of classification and labelling of chemicals
GRI	Global Reporting Initiative
Environmental Impact Indicators (EN 15804)	
GHG	Greenhouse gas
GWP	Global Warming Potential (kg CO_2 eq.)
GWP-fossil	Global Warming Potential from fossil sources (kg CO_2 eq.)
GWP-biogenic	Global Warming Potential from biogenic sources (kg CO_2 eq.)
GWP-luluc	Global Warming Potential from land use and land use change (kg CO_2 eq.)
GWP-total	Total Global Warming Potential (kg CO_2 eq.)
GWP-GHG	Global Warming Potential for greenhouse gases (kg CO_2 eq.)
ODP	Ozone Depletion Potential (kg CFC-11 eq.)
AP	Acidification Potential (mol H^+ eq.)
EP	Eutrophication Potential
EP-freshwater	Freshwater eutrophication potential (kg P eq.)
EP-marine	Marine eutrophication potential (kg N eq.)
EP-terrestrial	Terrestrial eutrophication potential (mol N eq.)
POCP	Photochemical Ozone Creation Potential (kg NMVOC eq.)
ADP	Abiotic Depletion Potential

ADP-minerals&metals	Abiotic depletion potential for non-fossil resources (kg Sb eq.)
ADP-fossil	Abiotic depletion potential for fossil resources (MJ)
WDP	Water Deprivation Potential (m^3)
Other environmental performance indicators	
PM	Particulate matter emissions (Disease incidence)
IRP	Ionising radiation, human health (kBq U235 eq.)
ETP-fw	Ecotoxicity, freshwater (CTUe)
HTP-c	Human toxicity, cancer effects (CTUh)
HTP-nc	Human toxicity, non-cancer effects (CTUh)
SQP	Land use related impacts / soil quality (Dimensionless)
Resource Use Indicators	
PERE	Use of renewable primary energy excluding renewable primary energy resources used as raw materials (MJ)
PERM	Use of renewable primary energy resources used as raw materials (MJ)
PERT	Total use of renewable primary energy resources (MJ)
PENRE	Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials (MJ)
PENRM	Use of non-renewable primary energy resources used as raw materials (MJ)
PENRT	Total use of non-renewable primary energy resources (MJ)
SM	Use of secondary material (kg)
RSF	Use of renewable secondary fuels (MJ)
NRSF	Use of non-renewable secondary fuels (MJ)
FW	Use of net fresh water (m^3)
Waste Indicators	
HW	Hazardous Waste (disposed) (kg)
NHW	Non-Hazardous Waste (disposed) (kg)
RW	Radioactive Waste (disposed) (kg)
Output Flow Indicators	
CFR	Components for Reuse (kg)
MR	Material for Recycling (kg)
MER	Materials for Energy Recovery (kg)
EEE	Exported Energy, Electricity (MJ)
EET	Exported Energy, Thermal (MJ)
Lifecycle Stages / Modules	
A1	Raw material supply
A2	Transport
A3	Manufacturing
A4	Transport to site
A5	Construction/Installation
B1	Use
B2	Maintenance
B3	Repair
B4	Replacement
B5	Refurbishment
B6	Operational energy use
B7	Operational water use
C1	Deconstruction/Demolition
C2	Transport to waste processing
C3	Waste processing
C4	Disposal
D	Reuse-Recovery-Recycling potential
Other Relevant Terms	
SVHC	Substances of Very High Concern
EC No.	European Community Number
CAS No.	Chemical Abstracts Service Number
MJ	Megajoule
kg	Kilogram
m^3	Cubic Meter

NMVOC	Non-Methane Volatile Organic Compounds
Sb eq.	Antimony Equivalents
P eq.	Phosphorus Equivalents
N eq.	Nitrogen Equivalents
CFC-11 eq.	Chlorofluorocarbon-11 Equivalents
CO_2 eq.	Carbon Dioxide Equivalents
kg C	Kilograms of Carbon
kg CO_2 eq.	Kilograms of Carbon Dioxide Equivalent
ND	Not Declared

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IEC 62941:2019 Terrestrial photovoltaic (PV) modules - Quality system for PV module manufacturing

ISO 9001:2015 Quality Management System

ISO 14001:2015 Environmental Management System

ISO 45001:2018 Occupational health and safety management systems

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VERSION HISTORY

This EPD is a new submission